An Explainable Data-Driven Platform for Optimal Wind Farm Siting

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1 Introduction

Wind farm site selection is a crucial yet highly complex challenge. Optimal siting involves much more than simply identifying windy locations; it requires balancing multiple, often competing factors including, but not limited to, proximity to the electricity grid, consistency of wind resources, land use constraints, environmental concerns, community acceptance, and economic viability [1–3]. The goal is to select locations that will not only generate the most reliable energy over time but also integrate well into the broader power system and surrounding landscape. However, this decision-making process is often not systematically optimized.

This issue has become even more pressing in the context of growing climate change awareness and the global transition toward sustainable energy systems. In Australia, policy commitments such as the Victorian Renewable Energy Targets, legislated in the *Renewable Energy Act (2017)* [4], reinforce this shift. These targets include:

- 40% of Victoria's electricity generation to be renewable by 2025
- 65% by 2030
- 95% by 2035

Meeting such goals will demand not only increased renewable energy capacity, but also smarter, more coordinated deployment strategies. This underscores the need for data-driven and holistic planning tools that can guide decision-making.

Currently, wind farm development in Australia is largely driven by a bottom-up planning approach. While site selection occurs within a defined legal framework and requires approval from energy market regulators, final decisions are often shaped by the interests of local stakeholders, like landowners, and by wind resource assessments conducted over relatively short timescales. Given that wind farms typically operate for 20 to 30 years, this approach can result in long-term inefficiencies [5]. Studies have shown that such a fragmented planning process can lead to suboptimal outcomes in terms of both total energy generation and supply stability. In contrast, a top-down planning model, where wind farm locations are determined with long-term demand patterns in mind, has demonstrated the potential for significantly greater efficiency and reliability [6].

1.1 Aim

This literature review aims to explore existing methods for wind farm site selection, both globally and within Australia. It will examine traditional approaches as well as recent techniques that incorporate artificial intelligence.

Findings from the review will inform the design of a user-friendly, explainable decision-support platform intended for a diverse range of users. These can include both technical actors such as infrastructure planners, power operators (e.g., AEMO, VicGrid) and decision makers, as well as ordinary citizens and local communities. In this context, clear and interactive representations of site selection criteria can help build trust, reduce misunderstandings, and support more inclusive planning. This is particularly important because the question of where to produce renewable electricity is often controversial, and the acceptance of new energy infrastructures, especially by the communities living near proposed sites, is critical, and it can lead to tension between stakeholders with different priorities or levels of technical understanding. [2, 7–9].

Figure 1: Recent protests against wind farms in Australia. **(Center)** A protest in Wollongong against plans for an offshore wind farm, May 2025 (Photo: Anna Warr, The Advocate). **(Right)** A rally on the front lawns of Parliament House in Canberra, February 24 (Photo: Mike Bowers, The Guardian). **(Left)** A sign photographed by the author in Warrnambool, March 2025

1.2 Document structure

Following the introduction, this report presents a structured literature review divided into several sections. The review explores the types of commonly used input data and critically examines traditional and AI based methods for wind farm site selection, with a particular focus on how explainable the methods are. It includes comparisons between approaches used across different countries, as well as a section dedicated to Australian visual and interactive tools, mainly focusing on mapping systems. These tools, coming from different domains than wind energy planning, offer inspiration and practical features that will help shape the design of the proposed platform. The State of the Art Summary then identifies gaps in the existing literature, particularly the lack of explainable and accessible visual tools that allow the creation of "what-if" scenarios. The Research Project Plan outlines a practical approach to addressing these gaps, including a section on the technical tools to be used, such as programming languages and frameworks. Finally, the Conclusion section highlights the key findings and provides a concise summary of the report.

2 Literature review

2.1 Data Inputs for Wind Farm Site Suitability Modeling

Before delving into the various methods used for wind farm site selection, it is important to first understand the types of data commonly employed in these studies. The selection of suitable wind farm sites is inherently data-driven, relying on a wide range of datasets to evaluate the feasibility, efficiency, and sustainability of potential locations. These datasets, which span technical, environmental, and social factors, are fundamental in determining the most appropriate sites for wind farm development. After reviewing numerous studies it becomes evident that the types of data used are similar, regardless of geographical location or the specific approach employed. The most common siting factors identified across these studies are summarized in the table below:

Siting Factor
Wind Speed
Distance to Protected or Wildlife Areas
Slope
Distance to Urban Centers
Distance to Roads
Distance to Transmission Lines or Substations
Life Cycle Costs
Distance to Airports
Distance to Water Bodies
Elevation
Distance to Animal Habitats or Migration Zones
Wind Power Density
Distance to Agricultural Areas
Flood Risk
Landslide Risk
Distance to Historic Places
Distance to Railroads
Distance to Military Zones

Table 1: Common siting factors used in (onshore) wind farm site selection

However, even with these shared factors, a significant challenge persists: the lack of standardization in how data is represented and utilized. As highlighted in [10], a consistent and standardized approach to data representation is absent across the studies. This variability in data handling can create discrepancies in the results and undermines the reliability of findings across different studies, especially if applied to the same region of interest. The wind speed, for example, may be categorized either as an economic factor [11, 12], or as a techincal factor [13, 14]; as a constraint [15] (setting a minimum threshold) or as an evaluation criterion [16] (taken into account in order to maximize potential energy output). Furthermore, the choice of data sources often remains unclear, with many studies failing to specify where their datasets were obtained [1, 13, 17–20], which hinders transparency and replicability in site selection research.

2.2 Methods

The methods used for wind farm site selection (WiFSS) have grown more sophisticated over the past decade, driven by the growing demand for renewable energy. In general, these methods can be grouped into two main categories:

- Multi Criteria Decision Making **(MCDM)** methods
- **Non-MCDM** methods

Each group including a range of different techniques, differing in how they handle and process data.

2.2.1 Multi Criteria Decision Making (MCDM) methods

MCDM has been an active area of research since the 1970s [21] and its methods are applied across a wide range of research fields, including mathematics, energy systems, computer science, and economics [22]. MCDM methods addresses decision problems involving multiple, often conflicting criteria, typically measured in incomparable units. MCDM methods are designed to support the evaluation and selection of alternatives in complex scenarios where trade-offs are inevitable [23], such as in wind farm site selection. Determining importance coefficients (weights) for each factor of the problem is the central challenge in MCDM methods.

Analytic Hierarchy Process (AHP) In the context of wind farm site selection, the Analytic Hierarchy Process is probably the most extensively applied MCDM method, with numerous studies employing it across different regions in England [24], Saudi Arabia [25], Ecuador [26], India [27], Spain [28], among others. AHP is a rank-order weighting method, which means that each factor is considered to have a different level of importance when determining the best location for a wind farm. This method is composed of two key stages: pairwise comparisons and weighted linear combination (WLC). The pairwise comparison stage is needed to assign relative importance weights to the factors of the problem (e.g. distance from electrical grid, averaged wind speed). While WLC is used in the final step to evaluate and rank each potential wind farm location based on the calculated weights [29]. For instance, one wind farm location may have a certain distance from a national park, a specific distance from the electrical grid, and a particular average wind speed. Another location, in turn, might have different values for these same factors. Experts, using their knowledge and judgment, compare the criteria by performing pairwise scenario comparisons. Their opinions are then used to generate a comparison matrix, which is subsequently used to establish the relative weights. These weights determine which factors are more or less important in the context of the overall goal of identifying the best location. The main advantage of this method lies in its ability to break down a large problem into smaller, more manageable subproblems (such as comparing pairs of scenarios at different hierarchical levels) [30]. On the other hand, a major challenge is the reliance on expert opinions, which can be difficult to obtain and select. Moreover, these opinions can be conflicting and uncertain, as they are based on human judgment, which inherently involves subjectivity [1]. To overcome this issue, such methods also embed the calculation of a consistency index, which serves as a proxy for the logical coherence of judgments and helps ensure that the decision making process remains as reliable as possible [23].

Fuzzy Analytic Hierarchy Process (FAHP) Despite the use of a consistency index, the traditional AHP methods still faces limitations in handling the ambiguity and vagueness inherent in human judgment. Experts evaluations are often imprecise or linguistically expressed (e.g., "slightly more important" or "being far from the electrical grid is much less favorable than being far from a national park"), which makes it challenging to make precise judgments in pairwise comparisons. To better capture this uncertainty, the Fuzzy Analytic Hierarchy Process (FAHP) extends the conventional AHP by incorporating fuzzy set theory [31]. In a notable application of these principles, a study focusing on on-shore wind farm site selection in South Korea [1] employed the FAHP method to determine factors weights more robustly. Similar approaches have also been used in other countries, such as Greece [32] and Nigeria [33].

Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) TOPSIS is a popular MCDM technique, originally introduced over four decades ago [34], and still remains (together with its variations) one of the most widely applied methods in multi-criteria analysis. The core idea is intuitive: the optimal alternative should be closest to the ideal solution, representing the best possible performance across all criteria, and farthest from the negative ideal solution, which represents the worst case scenario. By calculating the relative distances to these two reference points, TOPSIS provides a simple ranking of the alternatives. Its appeal lies in its simplicity and compatibility with a variety of decision making contexts. For example, in the context of wind farm site selection, the ideal location might correspond to high wind speed, a short distance to the grid, and low environmental impact. These criteria can be represented as a vector. Each site in the region of interest is also represented as a vector of values and then compared to the ideal and negative-ideal vectors. TOPSIS calculates the geometric distance of each site to both extremes and assigns a relative closeness score. The site with the highest score is considered the most suitable. TOPSIS allows for the inclusion of weights to reflect the relative importance of factors, adjusting the distance computations to the ideal and negative-ideal solutions. This is why techniques like AHP and TOPSIS are often used together. For example, in a study conducted in Turkey [35], a FAHP method was applied to determine the weights of the factors. These weights were then used in the TOPSIS framework to rank the alternatives. A similar idea was used in Eastern Macedonia and Thrace [36]. However, while TOPSIS offers an intuitive solution to the problem, it is not without limitations. A key concern arises during the data normalization phase, where vector representations are constructed. Since TOPSIS is highly sensitive to the way data is normalized, selecting an appropriate normalization method is critical. Poor choices at this stage can lead to distorted distances and, ultimately, unreliable rankings. This is particularly problematic when dealing with factors that have different units or highly skewed distributions. Recent studies have highlighted this issue [37, 38]. Additionally, TOPSIS during the distances caluclation assumes that criteria are statistically independent [39], overlooking potential interactions that may exist between them. Finally, the method relies on human input to define the ideal and negative ideal solutions, which introduces subjectivity into the process. As with Fuzzy AHP, fuzzy extensions of TOPSIS have also been developed to address this uncertainty [40].

The methods discussed so far, regardless of their specific type, commonly utilize Geographic Information Systems (GIS) to visualize the results. GIS refers to a system designed to capture, store, manipulate, analyze, manage, and present geographical data. In the context of wind farm site selection, GIS tools are essential for integrating various spatial datasets. These systems allow for the visualization of complex, multi dimensional data in a geographical context, making it easier to identify locations that are more suitable or less suitable. Each of the methods typically produces a ranking or a score for each location. This ranking or score can then be used as a form of legend on the GIS map. Several examples are provided in Figure 3.

2.2.2 Non-MCDM Methods

While MCDM methods offer structured frameworks for decision making, they rely heavily on subjective judgments to assign importance to the different factors. Making them inherently dependent on experts opinions and qualitative assessments. In contrast, Non-MCDM methods are predominantly fully data driven approaches. Although expert input may still be incorporated at certain stages, these methods fundamentally rely on computational models and algorithmic logic to derive conclusions. Within this broader category, AI models have emerged as a hot topic in recent years, due to their ability to handle complex, high-dimensional data and uncover patterns that traditional methods might miss. In the following section, we will discuss some examples of these Non-MCDM methods.

Cost/Benefit Models Unlike the MCDM methods described before, which emphasize a broader set of factors such as environmental, technical, and social criteria, cost/benefit models focus primarily on the economic aspects of wind farm site selection. These models are designed to evaluate the financial feasibility of different site options by com-

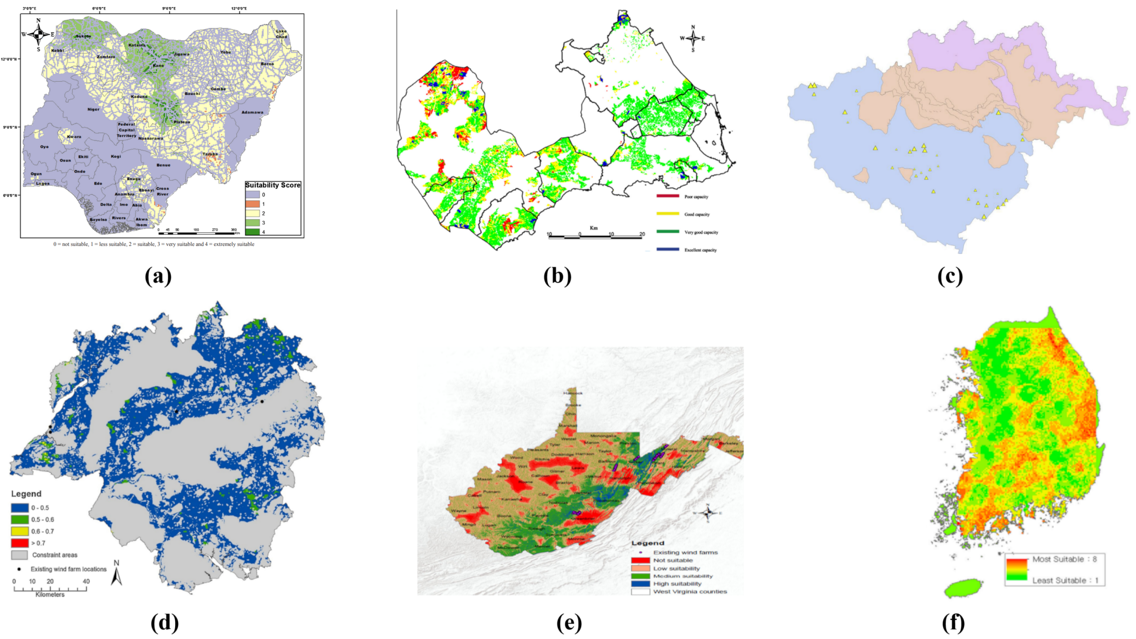


Figure 2: GIS-based visualizations showing the spatial distribution of site suitability for wind farm development. Regardless of the specific method used, each map represents suitability scores or rankings derived from multiple criteria.
(a) Nigeria, FAHP method [33] **(b)** Murcia region, Spain, FAHP method [28] **(c)** East Macedonia, Greece, AHP+TOPSIS method. Bigger triangles represent higher suitability [36] **(d)** South Central England, AHP method [16] **(e)** West Virginia, USA, AHP method [13] **(f)** South Korea, FAHP method [1]

paring the costs associated with wind farm development and operation against the anticipated economic benefits, primarily in terms of energy production and long term revenues. In the study by Kim et al. [41] for example, the cost estimation for offshore wind farms involves a comprehensive breakdown of all economic factors. Each cost category is meticulously calculated, including:

- Turbine cost
- Foundation cost
- Electric system
- Grid interface cost
- Project development cost

The benefit estimation is also detailed, taking into account the annual energy production and efficiency loss. The potential locations are then ranked based on the cost/benefit ratio and visualized using GIS. Such an approach leaves very little room for subjectivity, as it relies heavily on quantitative data and objective calculations. Similarly, a study focusing on Turkey's wind energy potential [42] applies a similar cost/benefit approach to evaluate optimal sites. What is particularly interesting in this study is their use of the ARIMA (AutoRegressive Integrated Moving Average) model to forecast where the highest energy demand will be in the future. By predicting energy consumption trends, the researchers are able to strategically place turbines closer to areas where energy demand is expected to peak. This ensures that the wind farms are aligned with future energy needs, making the entire planning process more forward looking and efficient. Also in this case, the estimation is entirely data-driven, as a mixed-integer nonlinear programming model is used to minimize installation, maintenance, and operational costs while accounting for wind potential, demand forecasts, and budget limits across the Turkish provinces.

Supervised Machine Learning Algorithms Machine learning algorithms represent a powerful evolution of data-driven wind farm site selection methods, offering predictive capabilities and advanced pattern recognition that go beyond traditional statistical or optimization models. Several supervised learning models have been explored in this context. For instance, in a study conducted in the Masovian Voivodeship of Poland, an Artificial Neural Network (Multilayer Perceptron) was trained to evaluate land suitability for wind farm placement using spatial and topographic features [43]. The training data was created by dividing the region into 100 m × 100 m land squares, each of which was evaluated by experts. Experts assessed each grid square and assigned a suitability score, which served as the target variable for training the model. The performance of the trained ANN showed strong predictive capability, with an accuracy value higher than 0.80 on the test set, indicating that the model was able to accurately capture the relationship between the input features and the land suitability. Similar good performance results were achieved using Support Vector Machines (SVM) and Random Forests (RF), as demonstrated by a study that employed global wind farm installation data [44]. Note that a common issue when using supervised learning methods is the imbalance in the dataset. In most cases, the labeled data, representing the actual locations where wind farms are placed or the optiaml ones, is significantly smaller than the set of all possible land locations. This class imbalance can cause the model to be biased toward predicting negative outcomes (unsuitable land). To address this issue, oversampling techniques like SMOTE (Synthetic Minority Over-sampling Technique) [45] are commonly used. SMOTE helps by generating synthetic samples for the underrepresented class, thus improving the model's ability to learn the characteristics of both classes more effectively and eventually obtain more accurate predictions. Other types of supervised models, such as Logistic Regression [46], are also used for similar purposes. However, we will not go into the specifics of these methods, as the underlying concept remains the same.

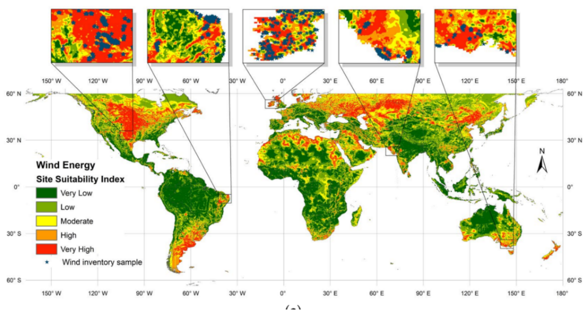


Figure 3: Global wind farm suitability map generated through supervised machine learning, trained on a dataset containing worldwide wind farm locations [44].

Unsupervised Machine Learning Algorithms Unsupervised Machine Learning offers an alternative approach to wind farm site selection that addresses some of the limitations associated with supervised learning models. Unsupervised techniques do not require ground truth data (labels). One critical issue with supervised learning in the context of our problem is the risk of model bias. When current wind farm locations are used as ground truth for training, the model basically learns to replicate the decisions made in the past, potentially reinforcing biases in site selection. This can result in the new model favoring areas similar to those where wind farms were previously established, even if those locations may not represent the most optimal set of choices. In other words, the model might automate a suboptimal decision-making process. If the supervised learning model is trained based on experts suggested locations (such as in [43]), instead of actual wind farm sites, it faces the same challenge previously discussed: human judgment. While experts can provide valuable input, their assessments can also be subjective.

The majority of unsupervised methods for wind farm location selection involve some form of clustering. A study demonstrated that Australian existing and planned wind farms produce less power and show greater energy output variability compared to a hypothetical optimal set of farms with the same capacity (even when the hypotetical set is constrained to locations within 100 km from the electrical grid) [6]. They arrived at this result by employing clustering technique and by minimizing correlations in wind power potential between different sites. They applied hierarchical clustering techniques to analyze historical wind speed time series data. What they discovered was significant: the existing and planned wind farms are largely concentrated within the same clusters. This means that these farms are highly correlated (i.e. if one zone experiences low wind conditions, other nearby farms will likely face the same issue). This interdependence creates a problem and it is particularly crucial when considering the need to complement solar energy, especially since solar power is unavailable at night and relies on wind or other sources for consistent, reliable energy production. Hierarchical clustering is not the only algorithm employed in these studies. K-Means and DBSCAN are also popular choices, as demonstrated by another Australian study [47] and in Pakistan [48].

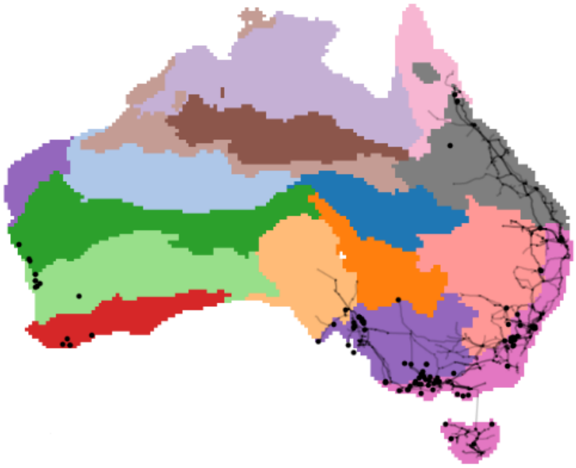


Figure 4: This figure illustrates the clustering of wind correlation zones based on historical wind speed data, showing that the majority of existing and planned wind farms (dots) are concentrated within a few key clusters. This concentration is understandable, as these regions align with the locations of the electrical grid (lines) and Australia's population centers. However, the analysis also reveals that there are also better and optimal sites for wind farms within a 100 km radius from the grid, suggesting potential for reducing interdependence among farms, which could imporve energy reliability. [6].

3 Summary of the State of the Art

Current research on wind farm site selection includes a wide variety of methods and approaches, particularly within Multi-Criteria Decision Making (MCDM) and AI-based techniques. Some of these were not even discussed in this document in order respect the assignment's length limits. However, in my opinion, a gap lies not in the availability of methods but in a critical lack of explainability and usability, especially from the perspective of non technical stakeholders such as citizens. This is not a secondary problem, because citizens are often directly impacted by the placement of new infrastructure, and tensions can arise if they do not fully understand the decision-making process or feel excluded from it. For instance, MCDM methods like AHP, FAHP, and TOPSIS are widely used and they heavily depend on expert derived factor weights and produce static GIS based maps. These outputs, while informative, do not support straightforward user interaction or allow non experts to explore alternative "what-if" scenarios based on different preferences or priorities. Few, if any, studies have incorporated dynamic interfaces that would allow users to adjust factor weights and immediately visualize how those changes affect site suitability outcomes. Introducing such tools could greatly increase both inclusivity and public trust. This is even more true for methods that fully rely on AI models and do not involve expert derived decisions. While explainability has been extensively studied in the field of Explainable AI (XAI), its application within the context of wind farm site selection is still relatively immature. Only a few recent papers, such as [49, 50], have begun to address this gap, and even then, the integration of explainability is still not presented in an interactive interface format. This is unfortunate, as GIS systems and XAI techniques are well suited for integration, as demonstrated in recent studies such as [51]. Moreover, while wind farm site selection has been extensively studied globally, Australia remains underrepresented in the literature. Wimhurst et al. in their comprehensive review of wind farm site suitability models up until 2022 [10], noted that no studies had specifically addressed this topic in the Australian context (Figure 5). Since then, a few papers have emerged, such as those referenced earlier (e.g., Rahimi et al. [47], Gunn et al. [6]), yet the volume of research remains limited. Especially given Australia's substantial wind resources, there is considerable potential for further exploration in this area.

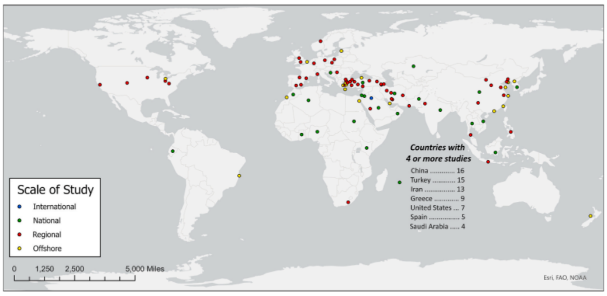
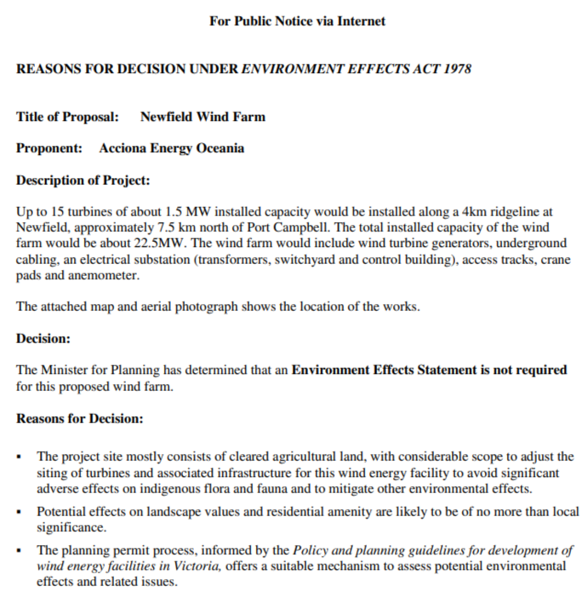


Figure 5: Study locations of articles on wind farm site suitability models published up to 2022 [10]. Note that there were no studies focused on Australia.

4 Research Project Plan

4.1 Overview

As discussed in the Summary of the State of the Art, in my opinion, there is a clear gap in the current body of research regarding the explainability and interactivity of wind farm site selection methods, especially from the perspective of non technical stakeholders. Although many advanced planning techniques exist, most of them are understandable only to experts with deep domain knowledge, making it hard for the general public to engage with or trust the outcomes. Thus, this research aims to help close that gap by developing a simple, interactive tool that makes wind farm site selection more transparent and easier to understand. The goal is to build a GIS based interface where users, whether they're citizens or planners, can explore possible locations, adjust the importance of different factors, and clearly see how those changes affect the results. Instead of creating a brand new method, the focus is on taking existing techniques and presenting them in a way that's more open and user friendly. In summary, the ultimate goal is to improve communication and understanding of how decisions are taken in wind farm planning, because very often they are not very well explained. To illustrate this point, the figures 7 and 6 below show two examples of wind farm developments in Victoria where the reasoning behind site selection could have been better communicated. And these aren't isolated cases. There are plenty more where the decision-making process could have been explained much more clearly. Such a tool could be used, for instance, alongside more lengthy government or industry documents like the Integrated System Plan For the National Electricity Market (2024) by AEMO [52]. To explain long-term planning, helping to break down complex information and make it more accessible.



For Public Notice via Internet

REASONS FOR DECISION UNDER *ENVIRONMENT EFFECTS ACT 1978*

Title of Proposal: **Newfield Wind Farm**

Proponent: **Acciona Energy Oceania**

Description of Project:

Up to 15 turbines of about 1.5 MW installed capacity would be installed along a 4km ridgeline at Newfield, approximately 7.5 km north of Port Campbell. The total installed capacity of the wind farm would be about 22.5MW. The wind farm would include wind turbine generators, underground cabling, an electrical substation (transformers, switchyard and control building), access tracks, crane pads and anemometer.

The attached map and aerial photograph shows the location of the works.

Decision:

The Minister for Planning has determined that an **Environment Effects Statement is not required** for this proposed wind farm.

Reasons for Decision:

- The project site mostly consists of cleared agricultural land, with considerable scope to adjust the siting of turbines and associated infrastructure for this wind energy facility to avoid significant adverse effects on indigenous flora and fauna and to mitigate other environmental effects.
- Potential effects on landscape values and residential amenity are likely to be of no more than local significance.
- The planning permit process, informed by the *Policy and planning guidelines for development of wind energy facilities in Victoria,* offers a suitable mechanism to assess potential environmental effects and related issues.

Figure 6: Reasons for the site selection of Newfield wind farm (Victoria), available to the public. Retrieved from: Newfield Wind Farm – Planning Referral.

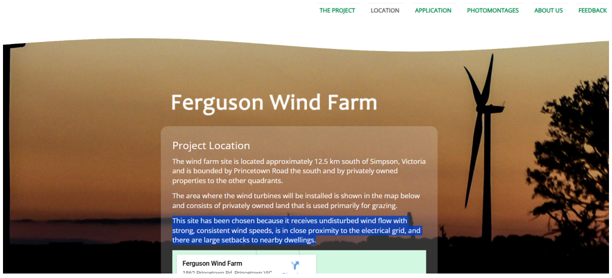


Figure 7: Ferguson Wind Farm – Project Location.

4.2 Plan

This project plan outlines a general approach to building the site selection tool. While the steps below are laid out as distinct phases for the sake of explanation and presentation, the process is intended to remain flexible. As the project develops, certain tasks will proabablyshift, overlap, or change based on what is learned in the way. The aim is to move forward in manageable stages, with space to experiment, gather feedback, and refine the tool.

- **Define the criteria and collect data**
  The first step is to decide which factors to include in the site selection process, guided by existing research and case studies. Once the criteria are finalized, the necessary spatial data, specific for Australia, will be gathered from open access sources. Probably the challenge at this point will be the harmonization of datasets from different providers, in terms of format, resolution, and coverage. This stage is very important to ensure a consistent foundation for the system. Since the primary focus of this project is on building a visual interface the number of criteria used at this stage may be fewer than in more comprehensive studies. However, the structure will remain open and extensible, allowing additional factors to be incorporated in the future. That said, several core criteria, such as wind speed, distance to the electricity grid, proximity to national parks, will definitely be included from the very beginning. Additionally, an extra consideration to increase feasibility and realism is narrowing the scope to focus exclusively on onshore turbines. Since the data used for this project is open source (e.g. ERA5 global reanalysis for weather data [53],electricity infrastructure locations) and does not involve human subjects or per-

sonal information, ethical concerns related to privacy or data protection are not needed.

- **Set up the basic GIS framework**
  Instead of using classical GIS tools, I plan to use web based mapping frameworks that are open-source and easy to deploy. This approach allows for greater flexibility and accessibility. Additionally, i'm planning to use Python for the backend of the interface, so that the system can can more smoothly integrate with various explainable XAI techniques. A good framework for building this setup is Plotly Dash, which provides both the interactivity needed for the frontend and the ease of use for the backend integration. The focus at this stage will be on loading and visualizing the prepared data for analysis. No logic will be implemented yet.

- **Add the logic**
  A simple decision-making method, such as a weighted overlay or AHP, to generate a suitability map.

- **Make it interactive**
  An interactive interface will be developed, allowing users to adjust weights and settings in real-time and instantly observe how the results change. This is the crucial functionality for making the tool both accessible and engaging. It is important to note that the optimal weights, typically required for techniques like TOPSIS or AHP, will not be calculated as part of this tool, as this would require expert judgment. Instead, the goal is to enable users to visualize how the suitability map changes based on any set of preferences or weights they choose, regardless of their optimality or accuracy.

- **Bring in explainabilty**
  This section can be viewed from two stages of explainability, each contributing to improving transparency.

  - **Improving the Weight-Based System**
    The weights based system previously implemented is inherently self-explanatory, as it allows users to adjust the importance of various factors and see the direct impact on the suitability map. To further enhance explainability, additional features can be integrated. For instance, if a user clicks on a specific location on the map, the interface could show how far that location is from the electricity grid (e.g., 300 km). It could then display how this 300 km compares to the distribution of distances for all points, providing users with context on whether this distance is relatively short, long, or within an expected range. The distance from the grid is used as an example here, but the same can be applied for all the different factors. For example, a very good illustration of this principle can be found in the Australian Cancer Atlas. In their interactive mapping tool, users can explore various health data layers and click on specific locations to view contextual information, such as how a particular area compares to the rest of the nation.
  - **Integrating an AI Model for Site Suitability**
    In this stage, a machine learning model will be implemented to generate an additional suitability map. While I do not expect the model to be perfect, especially given the challenges posed by a very imbalanced dataset (with only 100-200 actual or planned wind farm locations compared to 12,000 possible onshore locations across Australia, when for example gridded at 0.25° latitude-longitude resolution), the focus here is to explore the potential for integrating explainability into the model. The suitability map can be enriched with explanations regarding the decision-making process. For a decision tree, these explanations could, for instance, include Gini feature importances. For more complex models, such as ensemble methods or neural networks, techniques like LIME [54] or SHAP [55] can be used to offer interpretable insights into the model's decisions. The aim is to demonstrate that, independently of achieving high model performance, there is value in visually presenting explainability.

- **Test with users**
  The tool will be shared with a small group of

non-expert users to observe how they interact with it—what's intuitive, what's confusing, and where things can be improved.

- **Refine and adjust**
  Based on feedback, both the interface and the underlying system will be tweaked to make the experience smoother and more understandable.

- **Write everything up**
  The project will wrap up with a full write-up of the approach, key findings, and areas for improvement in future versions.

Gap explainable ai Mostra foto della mappa fino al 2022, e poi un solo studio dopo il 2022 Non spiegabili, static maps, not what if scenarions easily done (it requeires new computation) No python integration for future models (climate change)

cita assolutamente il paper di xai per gis system

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